

Base GTM Engine - Data Points

Every data point, its source, role in the product, geography and preparation. **[VERIFY]** = check early in the weekend. Rows are shaded by category.

Category	Data point	Source	Used for	Level	Coverage	What it gives the rep	Preparation
Eligibility	Utility territory / deregulated area	HIFLD Electric Retail Service Territories	Eligibility score	Territory polygons	All US	Don't waste time where Base can't sell	Spatial join block groups to territory polygons
Eligibility	Base service areas & utility partners	Base website / ask Base	Eligibility score, display	City / utility	Texas (+ Illinois)	Which pitch: direct sale or utility program	Manual tag: direct / utility partner / not served
Fit (ACS)	% owner-occupied	Census ACS 5-year (Census API)	ML feature, display	Block group	All US	Renters can't buy	API pull, compute share
Fit (ACS)	% single-family detached	Census ACS 5-year	ML feature	Block group	All US	Apartments/condos are poor targets	API pull, compute share
Fit (ACS)	Median home value	Census ACS 5-year	ML feature, display	Block group	All US	Ability to pay	API pull; handle missing values
Fit (ACS)	Median household income	Census ACS 5-year	ML feature, income-only baseline	Block group	All US	Ability to pay	API pull; handle missing values
Fit (ACS)	Median year built	Census ACS 5-year	ML feature, display, installability proxy	Block group	All US	Older homes may need panel upgrades	API pull
Fit (ACS)	% households with someone 65+	Census ACS 5-year	ML feature, display	Block group	All US	Older households often value backup	API pull, compute share
Fit (ACS)	Years in home / year moved in	Census ACS 5-year	ML feature	Block group	All US	Long-term owners invest in their homes	API pull
Fit (ACS)	Home heating fuel (electric vs gas)	Census ACS 5-year	ML feature	Block group	All US	Electric-heat homes suffer in winter outages	API pull, compute share electric outages
Fit (ACS)	Working from home share	Census ACS 5-year	ML feature	Block group or tract	All US	Outage = lost income	API pull; copy tract value down if needed
Fit (ACS)	Mortgage status	Census ACS 5-year	ML feature	Block group	All US	Financial profile of owners	API pull, compute share
Fit (ACS)	Rooms / bedrooms	Census ACS 5-year	ML feature	Block group	All US	Home size proxy (usage)	API pull, median
Fit (ACS)	Household size, presence of children	Census ACS 5-year	ML feature	Block group	All US	Household needs	API pull
Fit (ACS)	Housing cost as % of income	Census ACS 5-year	ML feature	Block group	All US	Affordability of monthly fee	API pull
Fit (ACS)	Vacancy rate	Census ACS 5-year	ML feature	Block group	All US	Vacant homes don't buy	API pull, compute share
Geography	Housing density, distance to center	Computed from TIGER/Line 2020	ML feature	Block group	All US	Suburban vs urban patterns	Compute from polygons + housing counts
Home (parcel)	Home size (sq ft)	County appraisal districts	ML feature, installability	Parcel -> block group	Per county (formats differ)	Bigger homes, higher usage	Clean export, keep residential, median per block group
Home (parcel)	Lot size	County appraisal districts	Installability	Parcel -> block group	Per county	Space for the battery	Clean, aggregate
Home (parcel)	Property type	County appraisal districts	ML feature, installability	Parcel -> block group	Per county	Single-family vs attached	Map codes to categories
Home (parcel)	Homestead exemption flag	County appraisal districts	ML feature, installability	Parcel -> block group	Per county	True owner-occupancy	Share of parcels with exemption
Home (parcel)	Deed date (recent purchase)	County appraisal districts	Segment filter	Parcel -> block group	Per county, if included [VERIFY]	New owners are open to offers	Share sold in last 12 months
Home (parcel)	Building footprints	Microsoft Building Footprints	Installability (outdoor space)	Building -> block group	All US	Room near house for battery	Footprint vs parcel lines gap
Derived	Installable share	Computed (parcels + footprints + permits)	ML feature, score multiplier	Block group	Where parcel data exists	Skip areas failing photo review	% homes likely meeting Base requirements
Permits	Generator + battery installs	City permit open data	ML LABEL , display	Address -> block group	Permit cities only [VERIFY]	Neighbors already invest in backup	Parse descriptions (keywords + LLM), geocode, per 1,000 eligible homes
Permits	Solar installs	City permit open data	ML feature, segment, talk track	Address -> block group	Permit cities only	Panels shut off in outages; Base fixes it	Parse, geocode, per 1,000 homes
Permits	Electrical panel upgrades	City permit open data	Installability proxy	Address -> block group	Permit cities only	Homes ready for install	Parse, geocode, count
Permits	Base Power installs (contractor)	City permit open data	Validation, map layer	Address -> block group	Permit cities only [VERIFY]	Base already here (social proof)	Filter contractor = Base Power
Permits	Competitor installs (contractor)	City permit open data	Competition layer	Address -> block group	Permit cities only	Where competitors sell	Filter contractor names (Tesla, Generac...)
Permits	Withdrawn / expired applications	City permit open data	Exploratory signal	Address -> block group	Some cities	Intent without follow-through	Filter status field
Permits (fallback)	Existing solar installs	Google Project Sunroof	Fallback label or feature	Tract	All US (check recency)	-	Join by tract ID
Need	Historical outages (count, hours)	DOE EAGLE-I	Need score, display, talk track	County	All US	'Lost power 4 times last year'	Aggregate outage events per county-year
Need	Severe weather events	NOAA Storm Events Database	Need score, display	County / zone	All US	Evidence of recurring risk	Count freezes/storms per county, 5 yrs
Need	Days above 100F	NOAA climate data	Need score	Station -> county	All US	Summer grid stress	Nearest station per county

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Timing	Storm / freeze alerts	National Weather Service API	Timing boost, feed	Forecast zone -> county	All US	'Ice storm expected Thursday'	Match alerts to counties
Timing	Recent outage	Utility outage maps / EAGLE-I replay	Timing boost, feed	Varies by utility	Per utility (fragile)	'Power went out here last week'	Replay historic event for demo
Timing	ERCOT grid stress	ERCOT via gridstatus	Timing boost, feed	Statewide	ERCOT area	Grid tight this week	Flag conservation appeals / low reserves
Grid value	Trading value per battery	ERCOT prices via gridstatus	Manager view, revenue potential	Load zone	ERCOT area (not El Paso)	Where the fleet earns most	Simulate 1 cycle/day on 15-min prices
Output	Propensity	ML model (LightGBM)	Priority score, estimated buyers	Block group	All Texas	Right households here	Percentile rank within metro
Output	Need score	Formula	Priority score	Block group (county detail)	All Texas	Real outage pain	Percentile rank of outage hours + weather
Output	Untapped flag	Model vs actual permits	Map layer	Block group	Permit cities	Demand not yet served	Shrinkage / Poisson test
Output	Priority score	Formula	Map, feed	Block group	All Texas	Where to knock	Elig x Install x Prop ^{0.6} x Need ^{0.4} (x Timing)